Data Source

The dataset is originally sourced from Kaggle and hosted on GitHub.
Data is then fetched using the GitHub API for ingestion into the Bronze layer.

Why did I take this data source?

This project uses a widely adopted multi-table dataset from Kaggle, commonly used in data engineering and analytics projects.

- Contains multiple related tables (customers, products, calendar, sales, returns)
- Enables implementation of joins and lookup operations
- Suitable for building an end-to-end data pipeline
- Includes historical data (2015–2017) for time-based analysis

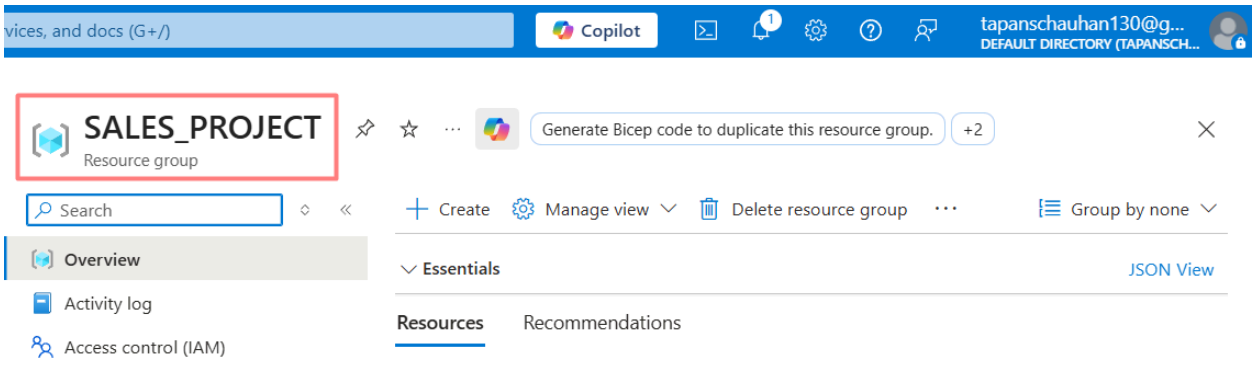
Key Tables

- **calendar.csv** → date and time-related data
- **customers.csv** → customer details (demographics, location, etc.)
- **products.csv** → product information
- **product_categories.csv / subcategories.csv** → product hierarchy
- **sales data (2015–2017)** → transactional records
- **returns.csv** → returned orders

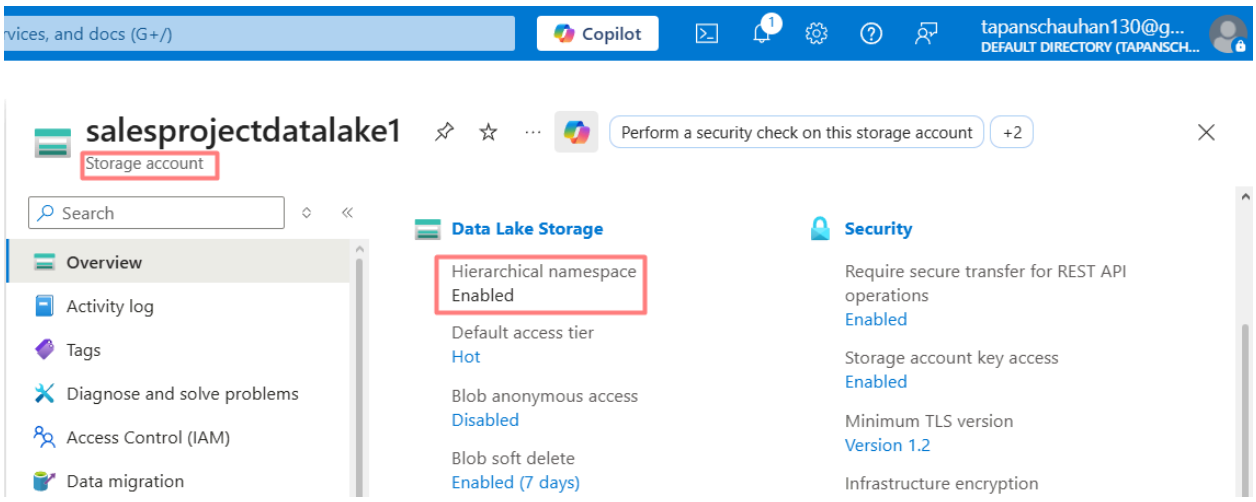
Summary

The dataset structure allows simulation of real-world scenarios where multiple data sources are integrated and processed using joins and transformations.

Create new resource group on Azure

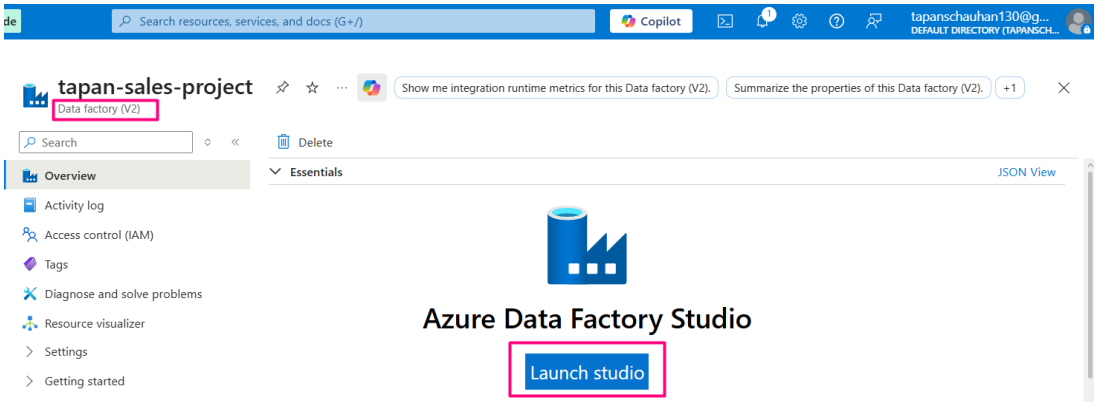


Create new storage account (resource)

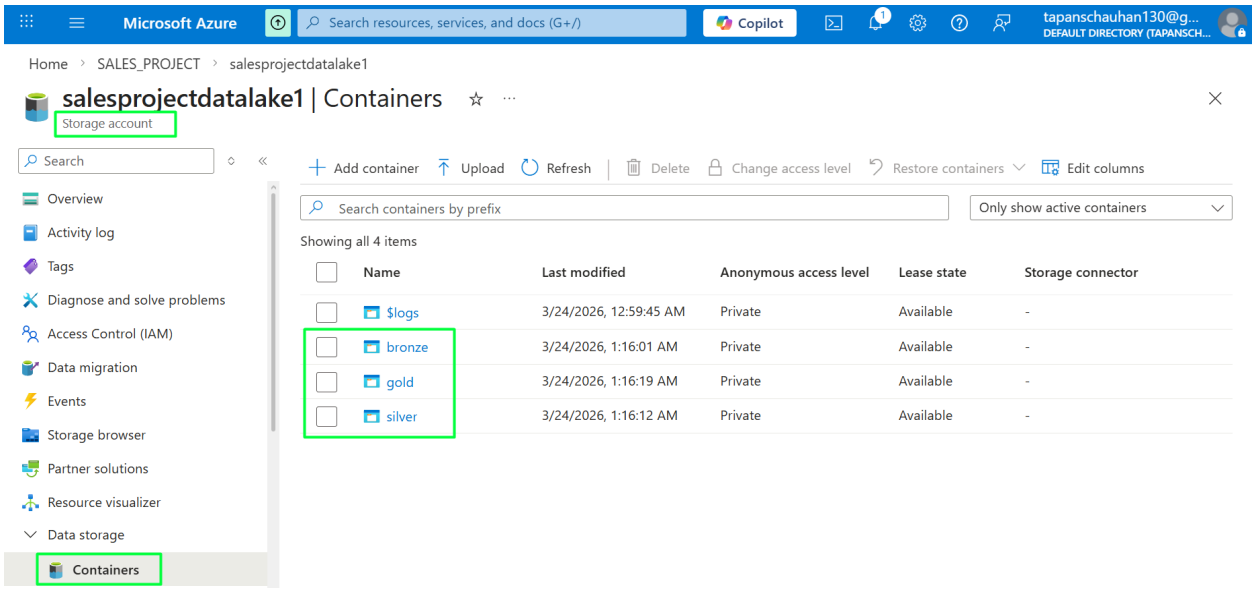


inside the resource group and enable hierarchical namespace it will create a data lake otherwise it will create a Blob storage.
data lake is better than blob storage because we can create a folder inside of folder in data lake.

Create a data factory

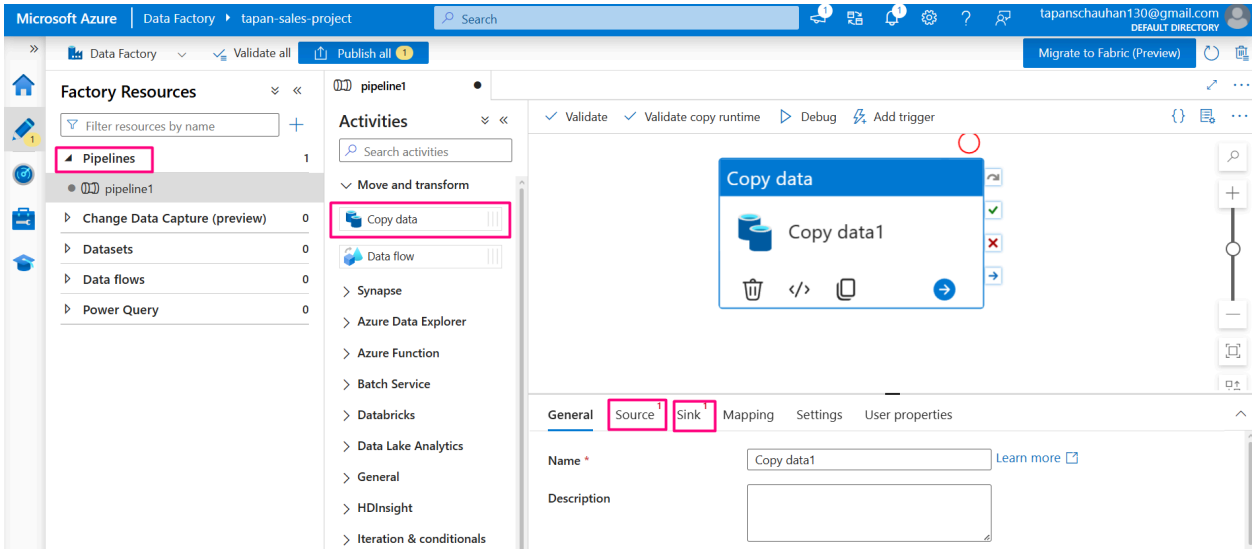


Now we will create containers for each layer
Click on storage account then in the data storage create containers

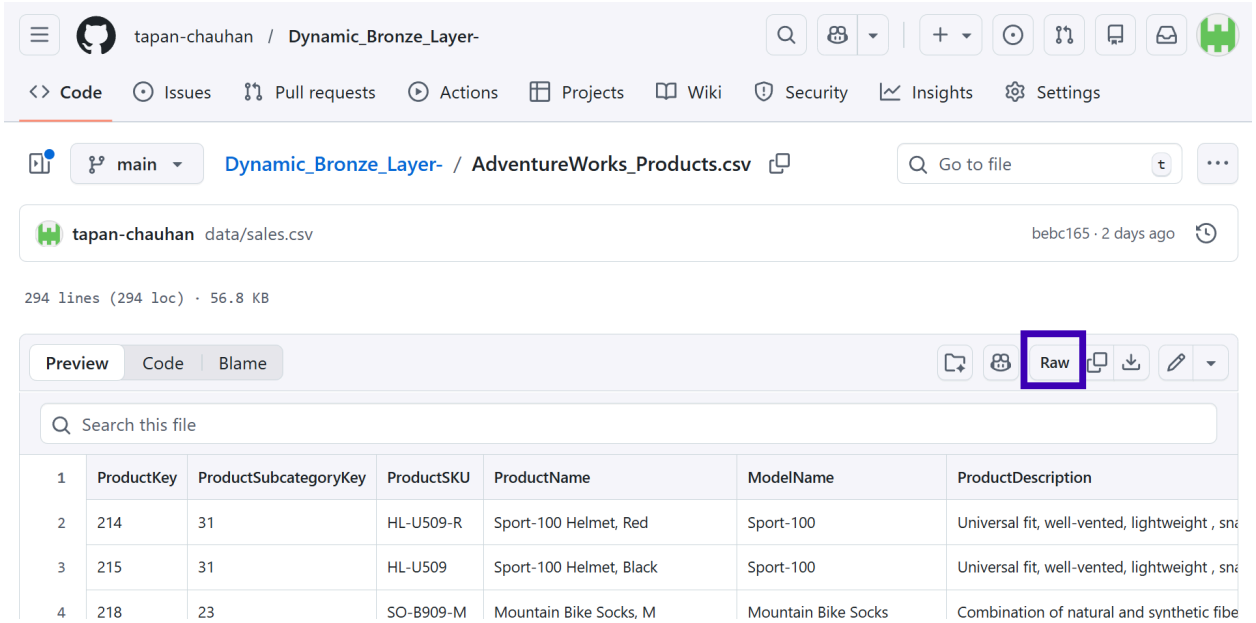


Click on data factory then data factory studio
Now we will load the data with the help of pipelines
To create a pipeline we need source and destination

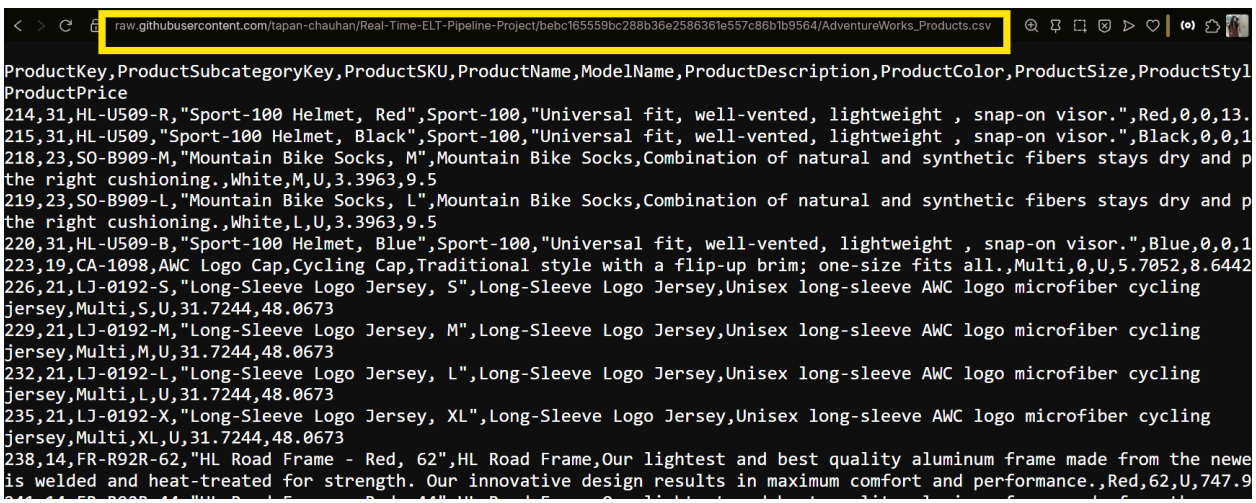
Click on pipeline then move and transform and then drag and drop the copy data into the work space then you will see source and sink



Now we will upload the kaggle data as csv file and then we will click on raw link



And we will use this URL to pull the data



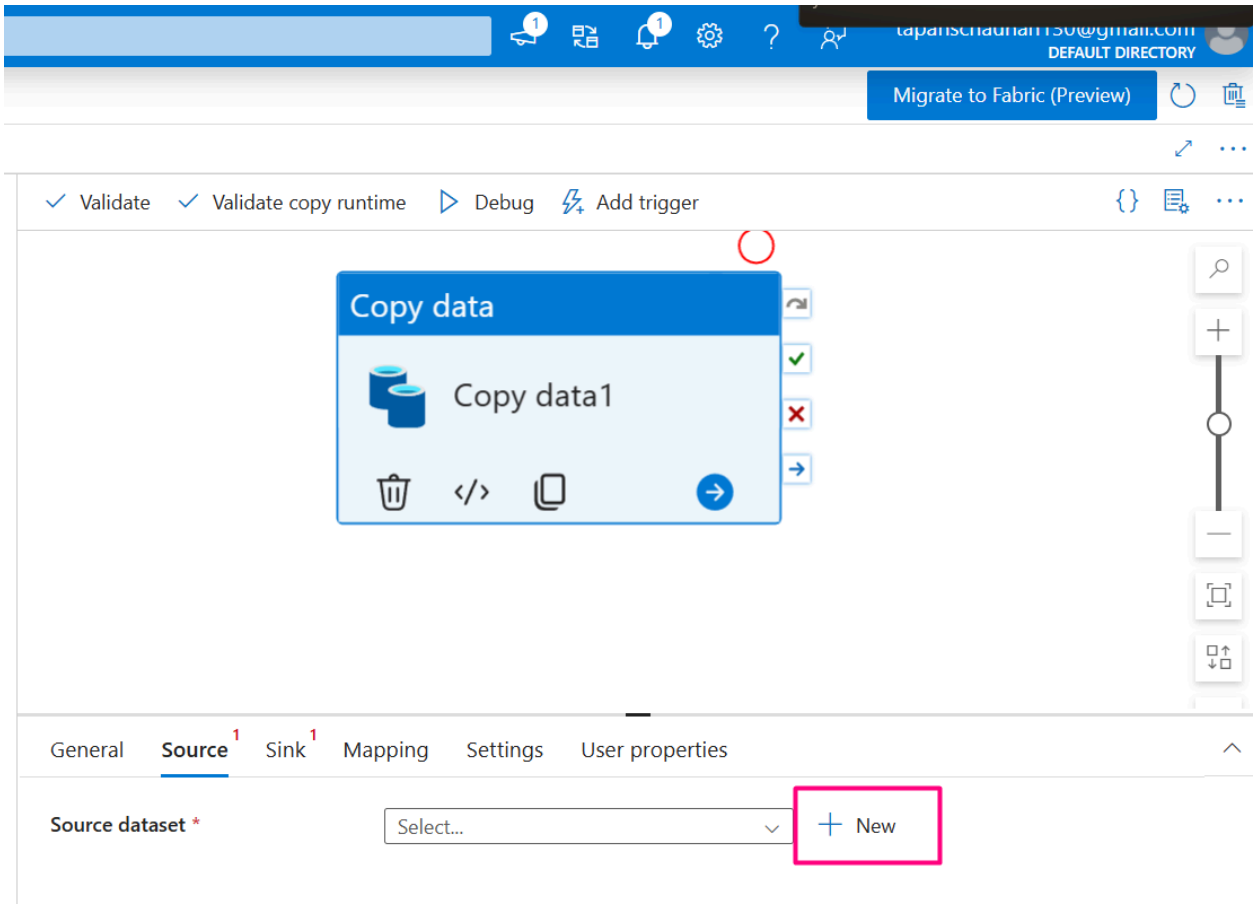
two parts base URL and relative URL

Now we will create connections these connections are known as Link services.

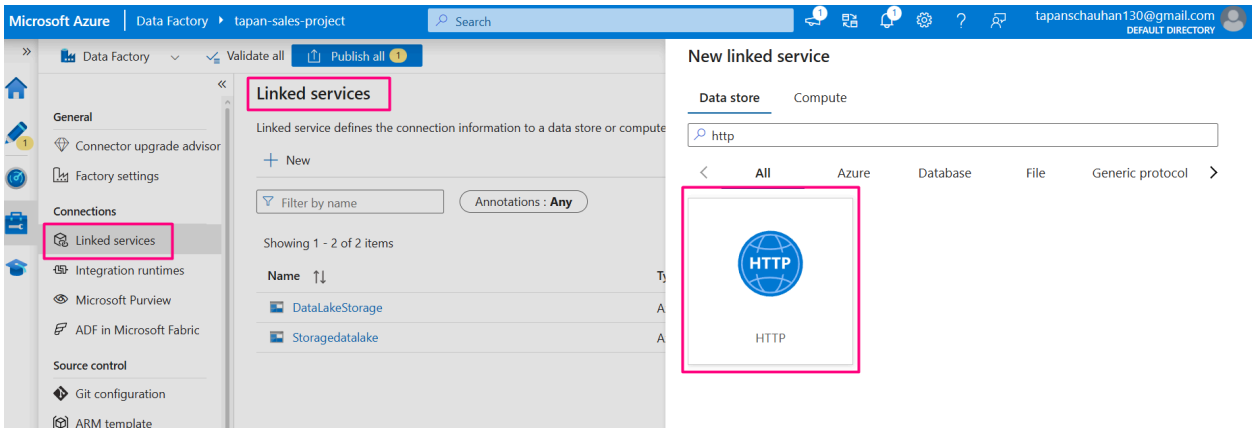
Now there are multiple files on get up we will take each file one by one and pull the date of from it

Take any file and click raw which has two parts base URL and relative URL

Now in the pipeline go to move and transform and drag and drop copy data now open source in create a new source.

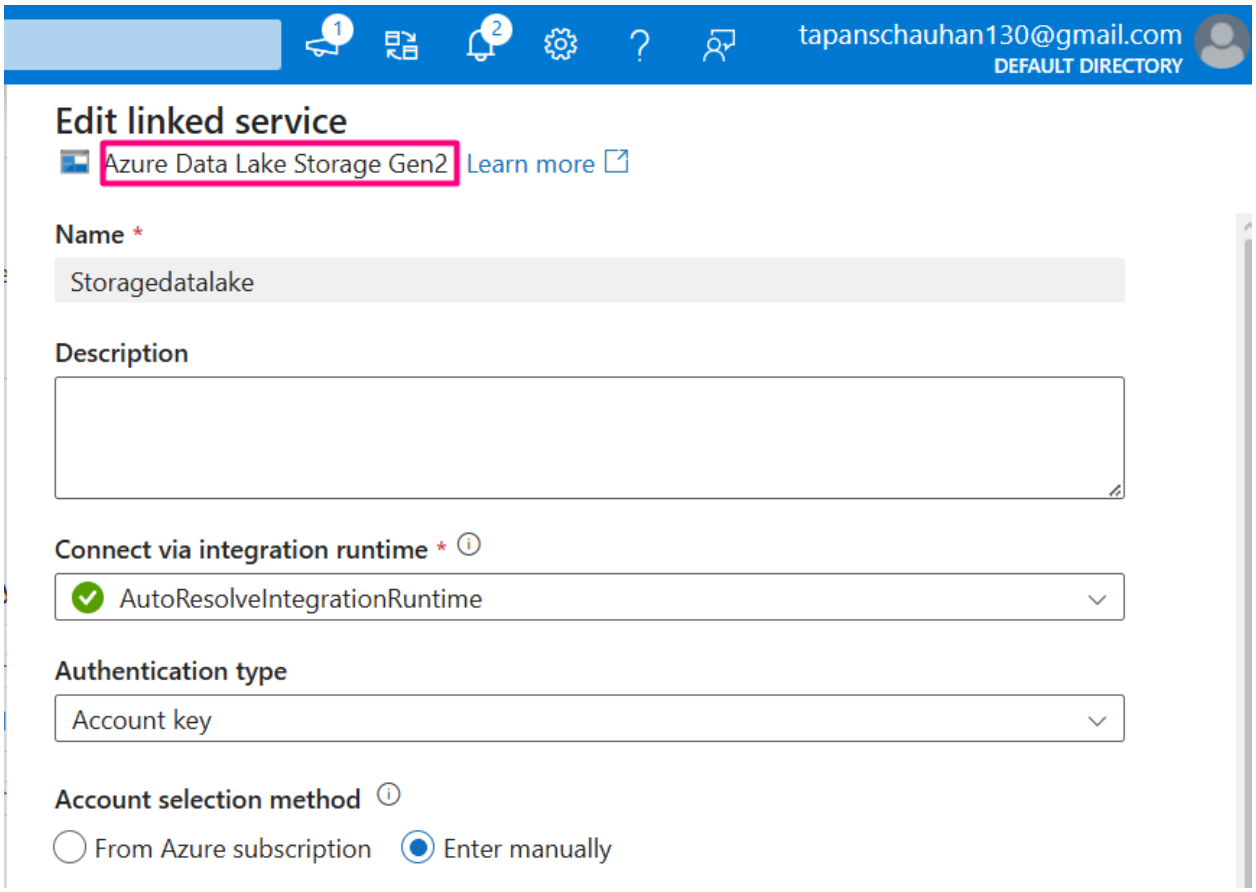


Go to manage tab click Link services create a new link service click on http and paste the base URL from GitHub

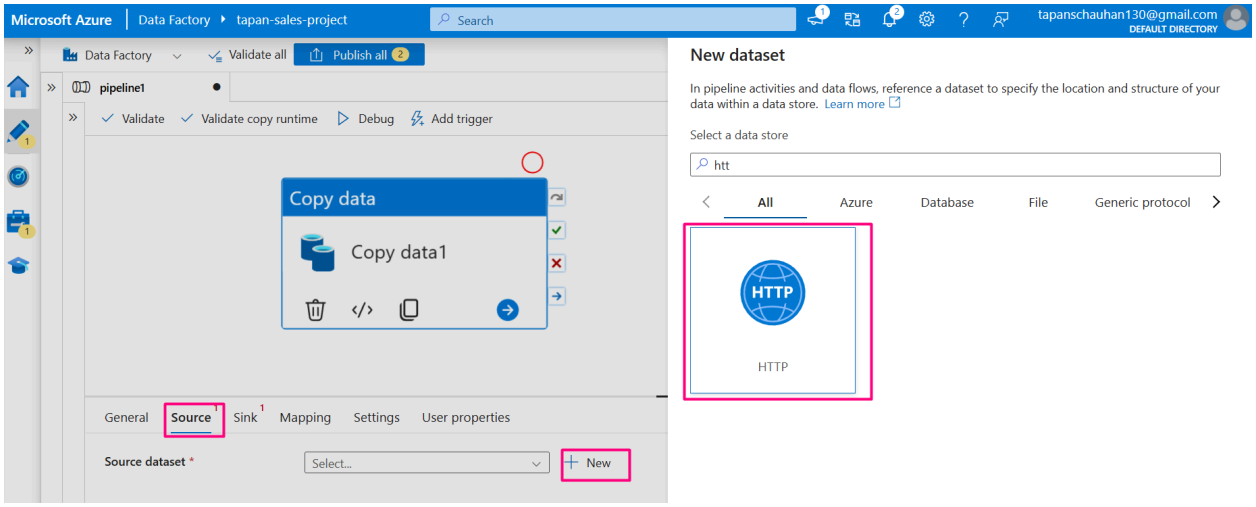


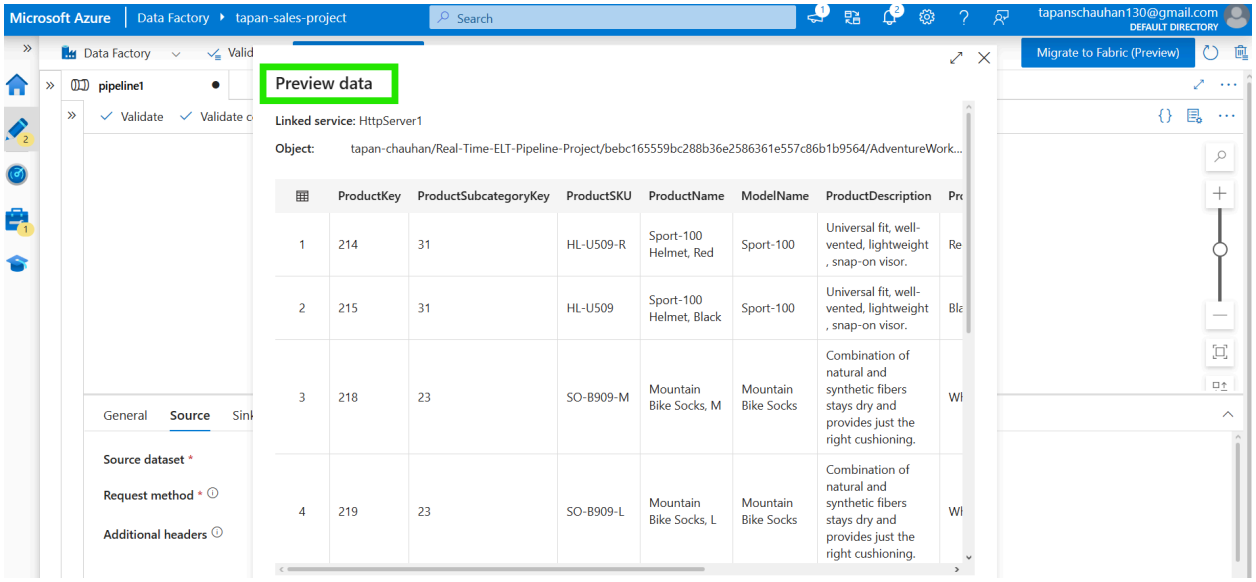
Server certificate validation enable and authentication type enormous create connection

Now create another link service with a data Lake storage generation 2 select the storage account name test connection and create the link service

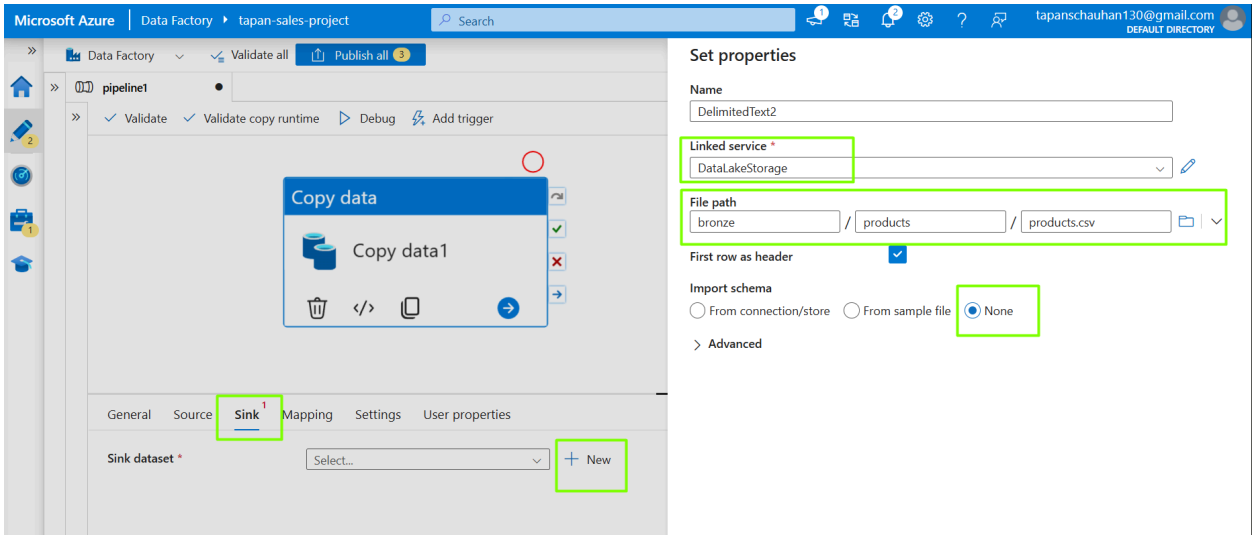


Now in the pipeline open source and create a new source and click http and select file type as csv and then select the link service and paste the relative URL of that csv file

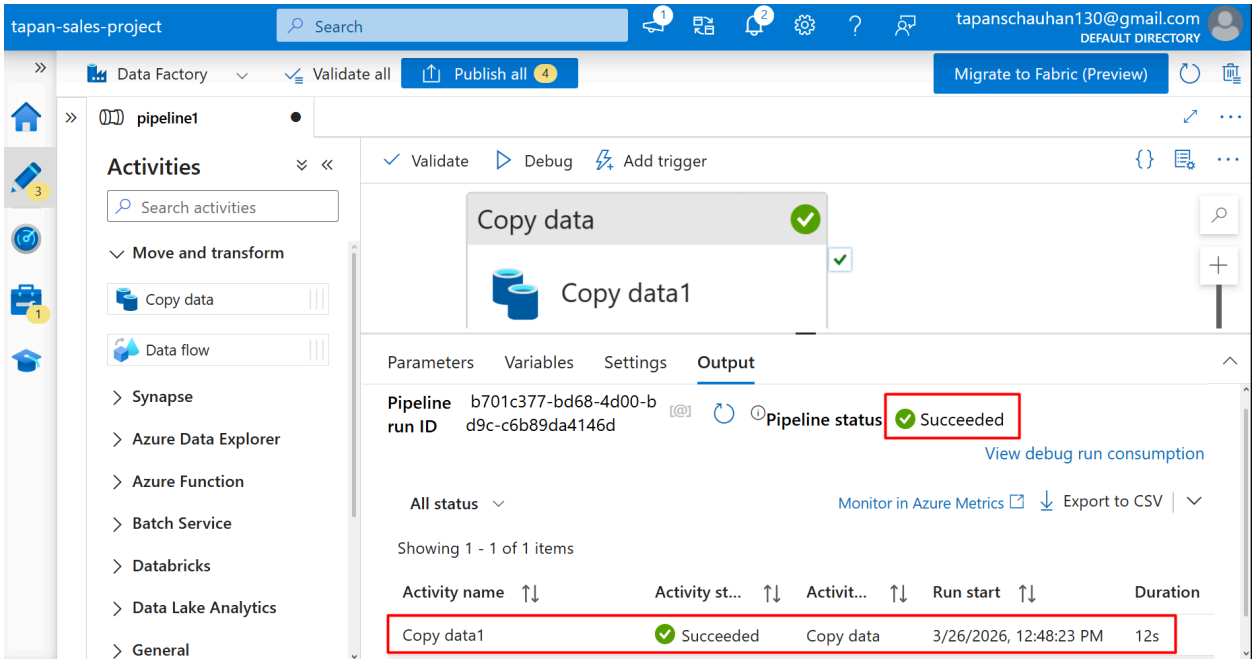




Now in sink create new click data lake and click csv now give the file path as bronze and create a new folder for that particular folder that you have copied from get up such as product click on none for import schema

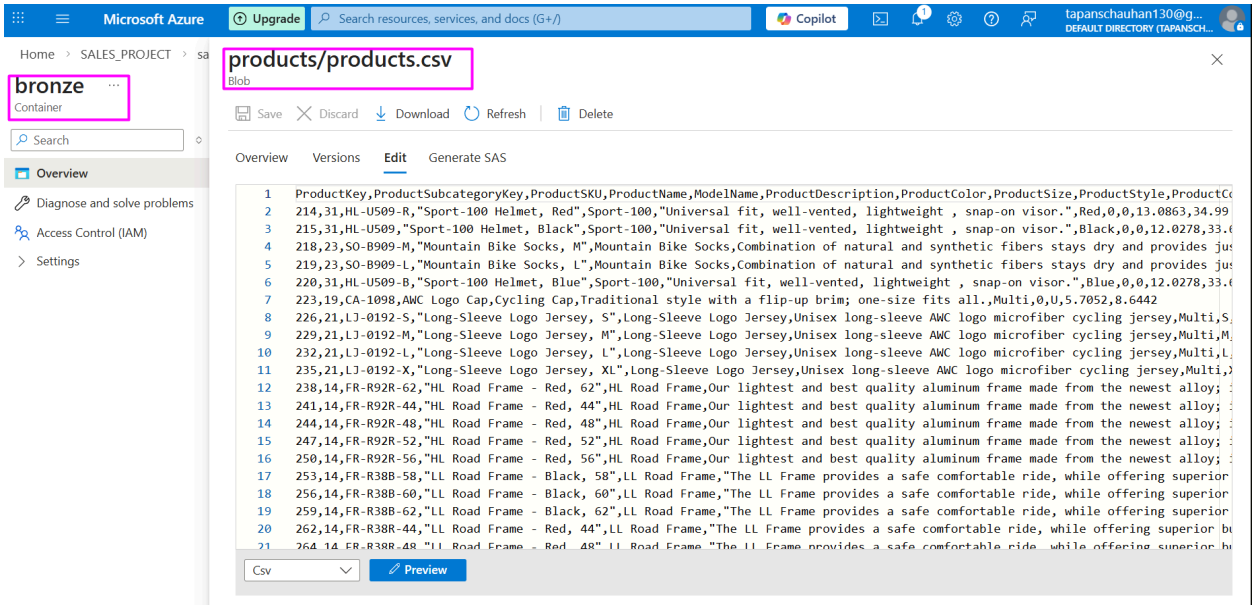


Now click debug

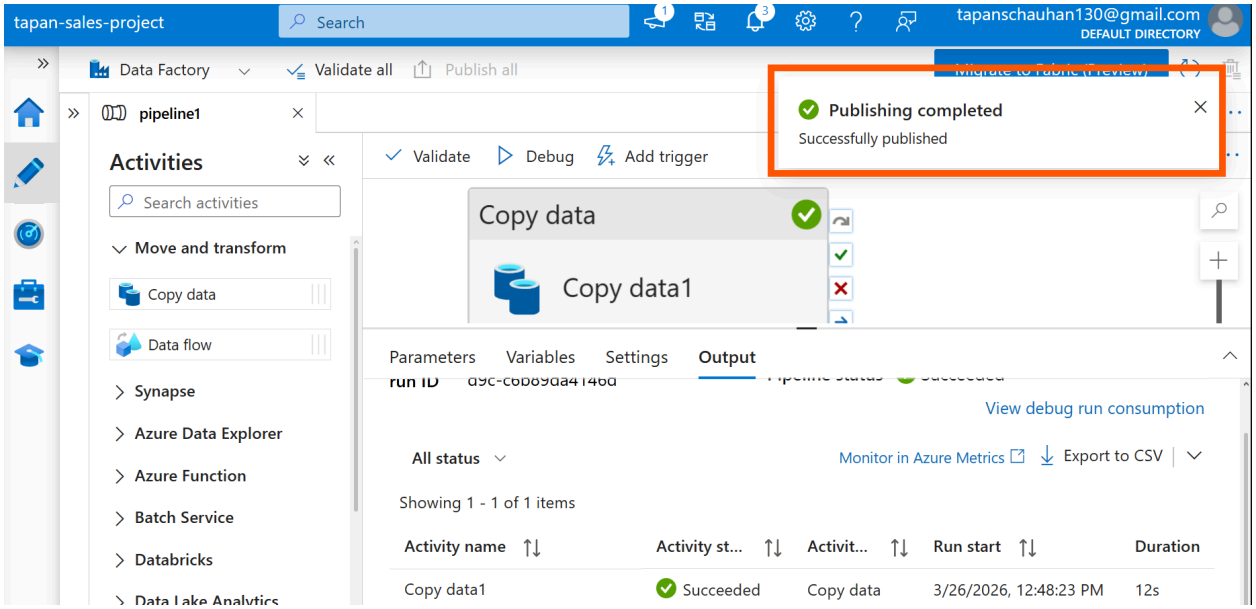


A data is loaded

We can see the data on our storage account inside the container bronze
we can see our loaded file there



Publish are pipeline to save



Now we have 8 to 10 files we cannot do this step manually so we will now built dynamic pipeline to do it automatically by it's just like applying loops

We will use iterations and conditions to perform dynamic pipeline.